

SURTHI SAMBHAV

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Education

KL University

Bachelor of Technology in Computer Science and Engineering | CGPA: 9.63

July 2023 – 2027

Vijayawada, Andhra Pradesh

Technical Skills

Programming: Python, Java (Basics), C/C++ (Fundamentals)

Data Analysis & Machine Learning: Pandas, NumPy, Scikit-learn, EDA, Data Cleaning, Feature Engineering, Model Training & Evaluation, Cross-Validation, Hyperparameter Tuning

Natural Language Processing (NLP): Tokenization, Stopword Removal, Stemming, Lemmatization, POS Tagging, NER, TF-IDF, Bag-of-Words, Word Embeddings

Deep Learning: PyTorch, ANN, CNN, RNN, LSTM, GRU, Transformers (Fundamentals)

MLOps (Basics): MLflow, DVC, Dagshub, AWS EC2, AWS S3, Docker, Git, GitHub

Web & Backend: HTML, CSS, TailwindCSS, React.js, Flask

Databases: MySQL, PostgreSQL, MongoDB

Projects

Financial Fraud Detection – Imbalanced Deep Learning System | GitHub

January 2026

- Built a fraud detection model on highly imbalanced transaction data (0.17% fraud cases) using a **Feedforward Deep Neural Network (Multilayer Perceptron)** and **Autoencoder-based anomaly detection**, achieving **0.96 ROC-AUC**.
- Applied advanced evaluation techniques including **Precision-Recall analysis, ROC curves, and threshold tuning** to minimize false negatives.
- Compared traditional ML models with deep learning approaches, improving fraud recall by **22%**.
- Integrated **MLflow** for experiment tracking and implemented a modular pipeline for reproducible training workflows.

Customer Churn Prediction – ML-Based Retention Analytics System | GitHub

December 2025

- Developed a predictive classification model using **Scikit-learn (Logistic Regression, Random Forest, XGBoost)**, achieving **87% accuracy and 0.89 ROC-AUC**.
- Performed extensive **EDA and feature engineering**, including tenure segmentation, contract-type encoding, and handled class imbalance using **SMOTE**.
- Optimized performance using **cross-validation and hyperparameter tuning**, improving churn recall by **18%**.
- Deployed the model via **AWS Elastic Beanstalk** with automated CI/CD integration using **AWS CodePipeline**.

Movie Recommendation System – Production-Ready ML Pipeline | GitHub | Live

November 2025

- Built a modular **Content-Based Recommender** using **TF-IDF, Word2Vec, and Cosine Similarity** on 45,000+ movies.
- Designed a production-grade architecture with separate **Data, Training, and Prediction pipelines** driven by centralized YAML configuration.
- Developed a scalable **FastAPI backend** integrated with the **TMDB API** for real-time metadata enrichment.
- Implemented custom logging, exception handling, and artifact management to demonstrate industry-ready ML engineering practices.

Certifications

- AWS Certified Cloud Practitioner (CLF-C02) | [Link](#)
- Microsoft Azure Fundamentals (AZ-900) | [Link](#)
- Cambridge Linguaskills - B2 | [Link](#)

Leadership & Extracurricular

Websico Club

Aug 2024 – Present

Director of Technology

KL University

- Led workshops for **30+ students**, improving coding proficiency and leadership skills.

VIDHTRA 2024 – Data Science & Ignition Event

Aug 2024

First Place

KL University

- Won **First Place** applying analytical reasoning to real-world data challenges.